June 19, 2026

Admissions Committee
Cranfield University
Bedford, UK

Dear Admissions Committee,

I am writing to express my strong interest in the PhD position regarding the development and optimization of ultra-efficient and power-dense ice protection systems for future aircraft. As an AI & Data Science Engineer with deep expertise in thermodynamic modeling and aerospace constraints, I am exceptionally well-prepared for this project.

For my State Engineer thesis, I developed a highly complex ”Gas Turbine Digital Twin.” This required rigorous modeling of thermodynamic degradation, temperature gradients, and fluid dynamics within high-stress mechanical environments. I engineered hybrid neural architectures (ST-KDFormer, Temporal Convolutional Networks) to predict thermal anomalies while strictly adhering to the laws of energy conservation. Furthermore, my work on Physics-Informed Neural Networks (PI-DeepONet) successfully embedded fluid dynamic equations (0D Windkessel) directly into the neural training loop via PyTorch Autograd.

This exact intersection of computational fluid/thermal dynamics and advanced AI optimization is critical for next-generation ice protection systems. I am highly motivated to apply my digital twin architecture experience to optimize power density, simulate thermal distributions, and design ultra-efficient de-icing control systems for aerospace applications.

I would welcome the opportunity to discuss my Gas Turbine Digital Twin research with you in an interview.

Sincerely,
Ismail Aissaoui
https://ismail-aissaoui.vercel.app